

Was Liberation a Delusion?

Liberation Day Tariff Shocks, Bilateral Trade Flows,
and Trade Diversion in 2025

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Abstract

On April 2, 2025, the United States called a new tariff shock “Liberation Day.” This paper asks whether the customs record can make that word believable. Executive Order 14257 announced very high reciprocal tariffs, but the rate that firms actually met at the border was not the same as the headline schedule. I build a monthly partner panel from January 2020 to January 2026 and measure the realized effective tariff rate as duties collected over import value, instead of treating announced rates as if they were all collected. The tariff shock was real, but it was smaller and more uneven than the political headline made it seem: the US-wide realized rate rose to about four times its pre-2025 average, while exemptions, USMCA treatment, and customs timing kept the collected burden far below the highest announced rate. US sourcing also moved, but mostly across foreign suppliers. China lost share, third-country suppliers gained, and the product-level overlap points more toward rerouting than toward production returning to the United States. In PPML gravity regressions with partner and month fixed effects, higher realized tariffs are associated with lower US imports, with the main coefficient on $\ln(1 + \text{ETR})$ equal to -1.68 . A market-share difference-in-differences gives the same direction. Therefore, the policy did change trade, but the change does not clearly match the promise of liberation. If liberation meant lower dependence and lower costs for American buyers, the customs ledger gives a much weaker answer than the slogan.

JEL Codes: F13, F14, C23.

1. Introduction

In 1958, Isaiah Berlin gave a lecture at Oxford that economists and political philosophers still debate, on whether humanity's craving for "liberty" had all along been pointed at the wrong idea. When asked what he meant, Berlin explained that there are two concepts of liberty. Negative liberty is having the freedom of being able to act on one's own wishes without any interruptions, which only involves your own demands without any third-party interference. The second, positive liberty, has a hidden flaw, since it describes itself harmlessly as the freedom to achieve one's greatest self, but gives no specific reference to what the word "greatest" means. From this, for thousands of years of humanity, people with the power to manipulate others have taken advantage of that flaw and have changed the meaning of "greatness of self" to suit their own needs throughout history, while claiming they are liberating those they rule (Berlin 1958). And similarly, for a day that claimed to free Americans of trade restrictions and international unfairness, on April 2, 2025, Donald Trump announced the "Liberation Day" of the United States against its world trade rivals, freeing the American economy. Or did he?

The reason for this word choice is not the topic of this paper, but it is worth highlighting because the results below test whether the American people were in fact made better off by the decisions made on that day. The answer may point in the opposite direction. With Executive Order 14257, the US government stated that the United States faced a national emergency over "the large and persistent annual goods trade deficits" and therefore used the International Emergency Economic Powers Act to impose sweeping tariffs on almost every US trading partner (Executive Office of the President 2025). If tariffs lower trade and pass the cost on to American buyers, the liberation framing starts looking more rhetorical than real.

Thus, the first question this paper tries to answer includes whether the Liberation Day shock actually decreased trade between the United States and its trading partners, or whether it only looked like a big change because the announcement itself was big. A fall in one month would not be enough to prove this, since trade moves all the time and can move for many reasons. Therefore, the effect has to be compared to what the literature already finds after tariff shocks, especially the short-run and long-run elasticities in Boehm, Levchenko, and Pandalai-Nayar (2022). My data only have ten months after the shock, so the size of the estimate has to be read with that short horizon in mind instead of being treated as a full long-run answer.

The paper also has to ask where the trade went if it moved, because the policy was not only described as a punishment against foreign unfairness, but also as a way to bring production back to the United States. However, if an American importer stopped buying a product from China and then bought the same HS6 product from another third-country supplier, this would not mean that production came back to the United States. It would only mean that the foreign source changed. In that case the country would not be free from foreign supply, but only from one foreign supplier, which is a very different kind of liberation than the one that was promised.

The treatment variable has to be built with that problem in mind as well, because a legal schedule is not automatically the burden that firms meet at the border, and using it alone would make the regression answer the wrong question. The realized ETR, measured as duties collected divided by general import value at the partner-month level, is therefore closer to the tariff that firms actually faced at customs.

The empirical method used for this is Poisson pseudo-maximum likelihood (PPML). The reason for this choice is explained fully in Section 3, but the short version is that trade data contain very large flows, very small flows, and zeros at the same time. PPML keeps those observations in the estimation instead of making the model depend on a log transformation that the data do not naturally fit.

The data are built from public sources so that the whole exercise can be checked again. The main inputs are monthly bilateral trade flows, customs duties and import values, tariff schedules, gravity controls, and product-level applied tariff data. Section 4 gives the full source list and explains how the pipeline joins them.

The closest research done on the same topic as the paper you are reading would be Fajgelbaum and Khandelwal (2026), since they also study the 2025 tariffs with monthly customs data and already show that the tariff was not mainly paid by the foreign exporter. The contribution here is narrower but more direct, because instead of only asking what happened to total US imports or total welfare, this paper asks which partners lost US market share, which ones gained it, and whether those gains look like real replacement or only rerouting around China. It separates the tariff that was announced from the tariff that was actually collected, and then follows how exports to the United States and imports from China changed after April 2, because if American buyers paid higher tariffs while the United States mostly changed which foreign suppliers it used, then the word “liberation” becomes much harder to defend with the data.

The rest of the paper follows that question in the order needed to answer it. Section 2 first makes clear what the policy shock actually was, because the tariff cannot be treated as one headline number when customs dates, pauses, carve-outs, and revised rates all changed the burden firms faced, and Section 3 then explains why gravity, PPML, and the difference-in-differences design are the tools used to separate the April 2025 shock from normal movements in trade. Section 4 builds the data piece by piece, so the reader can see where the trade flows, realized tariffs, gravity controls, and product-level measures come from, while Section 5 uses those data to move from the announced-versus-realized tariff gap to sourcing changes, product-level rerouting, and the regression estimates. Section 6 explains what these results can and cannot prove, especially because the time horizon is short, the policy was uncertain, and country-level averages hide product differences, before Section 7 returns to the word “liberation” and asks whether the evidence is strong enough to support that claim.

2. The Liberation Day Policy Shock

2.1 *What happened, and when*

April 2, 2025 was a Wednesday. By the end of that day, the United States had moved from a low-tariff economy to a country that was suddenly taxing imports from almost everywhere. The change was not only one number announced in one speech. It became a chain of dates, pauses, exemptions, and revisions. Customs does not collect a speech; it collects the rate that is legally active when the good enters. So the shock has to be read as a timeline, not only as a headline.

April 2 announcement. On April 2, President Trump signed Executive Order 14257 (Executive Office of the President 2025; Office of the Federal Register 2025a). The order used the International Emergency Economic Powers Act (IEEPA), which is a 1977 law that allows the president to regulate international commerce after declaring a national emergency. In this case, the emergency was described as “large and persistent annual goods trade deficits.” The order was published in the Federal Register on April 7. Its annex created a universal 10 percent ad valorem baseline on almost every trading partner, and then added higher reciprocal rates for named countries: 34 percent on China, 32 percent on Taiwan, 24 percent on Japan, 46 percent on Vietnam, and 20 percent on the European Union, among others (Cutler et al. 2025; National Taxpayers Union Foundation 2025). Canada and Mexico were not treated in the same schedule, because earlier fentanyl- and migration-related IEEPA orders already covered them. USMCA-compliant goods from those two countries therefore stayed exempt (Pillsbury Winthrop Shaw Pittman LLP 2025).

April 5 customs entry and the in-transit waiver. The 10 percent baseline did not take effect the moment the order was signed. It became active at customs entry on April 5. CBP guidance also created an in-transit exemption: goods loaded before 12:01 AM EST on April 5, and already moving toward the United States, were exempt from the 10 percent baseline (U.S. Customs and Border Protection 2025). Goods loaded between April 5 and April 9 paid the 10 percent baseline, but did not yet pay the higher country-specific rates. This detail helps explain why the collected tariff could not immediately equal the announced tariff.

Equity market reaction (April 3-7). Markets reacted as if the announcement was not already priced in. The S&P 500 fell 4.8 percent on April 3, its largest one-day decline since the beginning of the COVID-19 pandemic, and kept falling on April 4. Across two days, around \$4.7 trillion was erased from the S&P 500, another \$2 trillion from the Magnificent Seven, and around \$2.2 trillion from equity markets in the six most affected foreign partners (Cox 2025). This is not the main evidence of the thesis, but it is important context. If markets lose roughly \$9 trillion in two days, it is hard to argue that the tariff path had been calmly expected beforehand.

April 9 pause and China escalation. On April 9, the administration paused the higher country-specific rates for most partners outside China for 90 days. Those partners moved back to the 10 percent baseline. China did not. The US rate on Chinese imports was raised to 125 percent immediately, and China answered with an 84 percent tariff on US goods effective at midnight on April 10. Within days, the US-China rate reached 145 percent, which became the number most people remember from the policy. But even here, the headline number and the customs number were not the same.

April 12-14 technology carve-out. A few days later, the policy became narrower. CBP guidance published on April 13 and updated on April 14, retroactive to April 5, excluded a long list of consumer-electronics and capital-equipment products from the reciprocal tariffs. The list included smartphones, laptops and other computers, semiconductors, solar cells, flat panel TV displays, flash drives, memory cards and solid-state drives, routers, chipmaking equipment, and wireless earphones (U.S. Customs and Border Protection 2025). Most of HS Chapters 84, 85, and 90 were covered by this carve-out. That is not a small exemption, because those chapters include machinery, electrical equipment, optical instruments, medical instruments, and the biggest US import category from China.

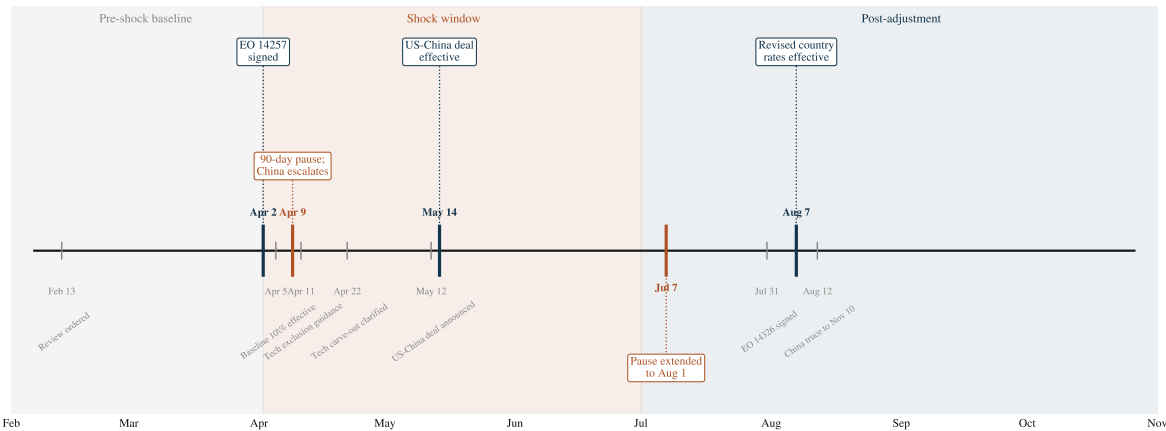
Geneva truce (May 11-14). On May 11, the United States and China announced a 90-day truce. The US tariff on Chinese imports fell from 145 percent to roughly 30 percent, while China’s tariff on US exports fell to 10 percent.

The truce became effective at customs entry on May 14 (York, Durante, and Watson 2026; National Taxpayers Union Foundation 2025). Because of that, the China series in the data does not stay at the April peak for long.

EO 14326 and revised country rates (July 31 onward). On July 31, the administration signed Executive Order 14326, changing the rates set under EO 14257. The revised schedule became effective at customs on August 7 (Office of the Federal Register 2025b). The China-specific truce was then extended to November 10 on August 12, and a narrower agricultural carve-out followed in November (National Taxpayers Union Foundation 2025). Across the whole period, pharmaceuticals (HS Chapter 30) and energy products (HS Chapter 27) stayed outside the reciprocal-tariff schedule.

Yale Budget Lab summary. By September 2025, Yale Budget Lab estimated that the average effective tariff rate faced by US consumers had reached 17.4 percent, or 16.4 percent after consumption shifts, the highest figure since the mid-1930s (Yale Budget Lab 2025a). This number is not the same as the customs-collected ETR used in this paper, because it includes every tariff action in force and not only the Liberation Day reciprocal schedule. Still, both measures point to the same problem: the tariff that was announced and the tariff that was actually collected are not identical.

Fig. I. Liberation Day Policy Timeline, 2025



Notes: The figure shows major announcement and customs-entry dates in the 2025 reciprocal-tariff sequence, set against three estimation windows: pre-shock baseline (gray), shock window (sienna), and post-adjustment (navy). Major labels mark the events used as identification anchors. Apr 2 marks the signing of Executive Order 14257; Apr 9 marks the 90-day pause for most partners and China's escalation; May 14 marks the effective date of the temporary US-China tariff reduction; Jul 7 marks the extension of the pause before its July 9 expiry; Aug 7 marks the effective date of the revised country-specific rates under EO 14326. Smaller tick marks below the spine record context dates: Feb 13 (reciprocal tariff review ordered), Apr 5 (10 percent baseline becomes effective at customs entry), Apr 11 (US Customs and Border Protection guidance on technology-sector exclusions), Apr 22 (electronics carve-out clarified), May 12 (US-China deal announced), Jul 31 (EO 14326 signed), and Aug 12 (China-specific suspension extended to November 10). Source: author's compilation from the Federal Register, the White House, and US Customs and Border Protection.

Figure I puts these dates on one line, because the policy is much easier to understand when the announcement dates and the customs entry dates are separated.

2.2 Why the shock is useful, and where the variation comes from

The reason this shock is useful for estimation is not only that it was large. It is useful because it was large in different ways for different countries, in different months, and for different products. It also came as an executive-order shock rather than as a long legislative process, and the two-day equity-market reaction suggests that markets had not fully priced it in beforehand (Cox 2025; Pattararangrong and Suwanprasert 2025). This gives the paper the kind of variation

it needs: some partners were hit hard, some were barely hit, and some were hit on paper but then partly protected by exemptions.

Across countries. The April 2 schedule created very different intended rates across partners. USMCA-compliant goods could remain at zero, many partners faced only the 10 percent baseline, the European Union faced 20 percent, Japan 24 percent, Taiwan 32 percent, China 34 percent at the first stage, Vietnam 46 percent, and a long tail of smaller partners faced still higher rates (Cutler et al. 2025; National Taxpayers Union Foundation 2025). By September 2025, the country-specific applied tariff built from the Teti database in Section 3.7 reaches 59.3 percent for China and only 4.1 percent for Ireland, with the United Kingdom at 9.0 percent. This spread from near zero to almost sixty percent is the main difference across countries that the regression uses.

Across months. The same timeline also changed treatment over time. Some dates changed the rate for almost everyone, while others only changed the burden for China or for exempt product groups. The month-year fixed effect in equation (3) absorbs what happens to everyone in the same month. What remains is the movement of each partner around those policy dates.

Across products. The April technology carve-out also means that countries with similar total exports to the United States did not necessarily face similar tariff burdens. Japan ships heavily in HS 84, 85, and 90, which were largely carved out, so its September 2025 product-weighted tariff is 19.8 percent. Vietnam ships more apparel and furniture, which stayed tariffed, so it sits at 21.6 percent even though its headline schedule was lower. Japan therefore paid more on paper and less in practice, while Vietnam paid less on paper and more in practice. The country-specific tariff measure in Section 3.7 uses each partner's pre-shock product mix to bring this product-level difference back into the country-month panel.

2.3 The anticipation threat

The main threat is anticipation. If firms knew the tariff was coming and sent goods early, then the pre-shock months would look artificially high, and the fall after April 2 would look too large. This is the cleanest way the estimate could exaggerate the tariff effect.

This concern still matters, but it does not remove the comparison. The full schedule was not publicly known far in advance, and the customs date was April 5, while the in-transit waiver protected goods that were already loaded before that date. A shipment already on its way did not have to be rushed in the last few days before the announcement to avoid the baseline rate.

Pattararangrong and Suwanprasert (2025) do find daily front-loading in Thai exports after the announcement, especially in agricultural products, but that is not the same as a pre-April 2 surge. Their pattern is more like adjustment inside the shock window, and it fades within weeks.

The pre-trend test in Section 5 checks the same problem directly. It asks whether partners that later received higher tariff exposure were already moving differently before April 2. They were not: the interaction of treatment intensity with the pre-shock time trend is basically zero ($\hat{\beta}_{\text{pre}} = -0.0011$, $p = 0.64$). This does not make the design perfect, but it does remove the most obvious pre-existing-drift story.

3. Theory and Identification

The empirical strategy has to do three things at the same time. It has to compare trade across countries that were hit by different tariff rates, it has to deal with the fact that trade data are very uneven and include zeros, and it has to separate the April 2025 tariff shock from everything else happening in the world economy. For that reason I use a gravity equation, estimate it with Poisson pseudo maximum likelihood (PPML), and then use a difference-in-differences design around the April 2 cutoff. This section explains why those choices are used and what exactly is being estimated.

3.1 What this paper estimates

The object I estimate is the short-run import response to the tariff shock. Put more simply, if a partner's realized effective tariff rises by one percentage point, how much less does the United States buy from that partner, after holding fixed the partner's normal level of trade and the global conditions of that month? This is not asking whether tariffs are good or bad in every possible model. It is asking what happened to US-bound imports in this data after the tariff actually hit customs.

The word short-run matters here. Boehm, Levchenko, and Pandalai-Nayar (2022) estimate an impact response of about -0.26 , a year-one response near -0.76 , and a long-run response between -1.75 and -2.25 after seven to ten years. My panel only runs ten months after the shock. So I am not estimating the full long-run elasticity, and I am not building a complete welfare model with counterfactual trade flows. Fajgelbaum and Khandelwal (2026) do that kind of calibrated structural work for the same shock. What I estimate here is more direct: a short-run elasticity, a market-share diversion effect, and a classification of which countries look like they gained or lost from rerouting.

3.2 Why gravity is the right setup

Tinbergen (1962) noticed that trade behaves like gravity: big economies trade more with each other, and distance pulls trade down. For a long time this was mostly an empirical pattern, not a fully grounded theory. The later structural gravity literature made the same pattern usable for estimation, because it showed what the variables in the gravity equation actually stand for.

Anderson (2011) shows that a gravity equation comes out of a trade model with constant elasticity of substitution (CES) demand, one industry, and iceberg trade costs. The equation is

$$X_{ij} = \frac{E_j Y_i}{Y} \left(\frac{t_{ij}}{P_j \Pi_i} \right)^{1-\sigma}, \quad (1)$$

where X_{ij} is exports from i to j , E_j is importer j 's expenditure, Y_i is exporter i 's income, Y is world income, and t_{ij} is the bilateral trade cost. The terms P_j and Π_i are called multilateral resistance. They cannot be ignored, because a country does not only decide whether to trade with one partner or not; it compares all possible partners at the same time. If every alternative supplier becomes expensive, then one bilateral relationship can look stronger even if nothing about that pair changed. The elasticity σ measures substitution across goods.

For that reason, fixed effects are not just a technical addition. In the specification below, partner fixed effects absorb the things that do not change for a partner, such as distance, language, contiguity, colonial links, baseline FTA status, and the outward resistance term Π_i . Month-year fixed effects absorb shocks common to all partners in the same month, such as the dollar, oil prices, the global business cycle, and the time-varying part of inward resistance P_j . After those two absorptions, the coefficient on $\ln(1 + \text{ETR}_{c,t})$ is identified from the tariff variation that remains inside the partner-month panel.

The problem with ignoring this can be seen from Anderson and Wincoop (2003). If multilateral resistance is ignored, gravity estimates become badly biased. The famous McCallum (1995) result found that Canadian provinces traded

twenty-two times more with each other than with similar US states. Without multilateral resistance, this looked like a huge border effect. Once resistance was modelled correctly, the implied border friction fell to 44 percent. The lesson is that trade with one partner cannot be understood without knowing how hard it is to trade with all other partners.

Eaton and Kortum (2002) reach a similar gravity form from a Ricardian model. In their setup, productivity for each good in each country is drawn from a Fréchet distribution with location T_i and shape θ . The probability that one country is the cheapest supplier to another becomes another version of gravity, where T_i acts like economic mass and θ becomes the elasticity. Their baseline estimate, $\theta = 8.28$, is a long-run trade elasticity in that world. Carrere, Mrazova, and Neary (2020) describe the same broad rules across gravity models: trade rises with mass, falls with distance, and shows home bias.

For panel data, Yotov et al. (2016) turn these ideas into practical rules: use panel data, absorb multilateral resistance with fixed effects, control for pair-level endogeneity where possible, and estimate with PPML. Baldwin and Taglioni (2006) call ignoring multilateral resistance the gold-medal mistake in gravity estimation. This paper follows that logic: month-year fixed effects absorb common time shocks and the time-varying part of resistance, while partner fixed effects absorb each partner's time-invariant position relative to the United States.

3.3 Why PPML, not log-linearised OLS

Before Santos Silva and Tenreyro (2006), the common way to estimate gravity was to take logs and run OLS. The problem is that trade data do not behave in the clean way this method needs. Large flows have larger errors, small flows have smaller errors, and the variance changes with the level of trade. In that case, logging the dependent variable does not simply make the model nicer. It changes what is being estimated and creates bias.

The technical reason is Jensen's inequality. Log-linear OLS estimates $\mathbb{E}[\log X | x]$, while the gravity model is about $\mathbb{E}[X | x]$. These two are not the same when the error is heteroskedastic. The gap between them moves with the regressors, and that movement enters the coefficient. A larger sample does not solve this, because the problem is in the transformation itself.

Zeros create a second problem. About 15 percent of the country-month observations in this panel have zero or near-zero trade. A log model cannot take $\log 0$. Dropping those observations would mean deleting the cases where trade is weakest, and adding a small constant would make the answer depend on an arbitrary choice.

PPML deals with both problems. The estimator assumes

$$\mathbb{E}[X_{ij} | \mathbf{x}_{ij}] = \exp(\mathbf{x}'_{ij}\beta), \quad (2)$$

and only needs the conditional mean to be correctly specified. It does not require the data to literally follow a Poisson distribution, and it does not require constant variance or normal errors. Zeros can stay in the sample because zero is a possible outcome under an exponential mean. In Santos Silva and Tenreyro (2006), the distance elasticity moves from -1.17 under log-OLS to -0.78 under PPML, which is a large correction for a coefficient that gravity models had already been using for decades. Fally (2015) also shows that PPML with importer-time and exporter-time fixed effects is numerically equivalent to solving the Anderson-van Wincoop structural model with multilateral resistance. This is why PPML is used here.

3.4 Fixed effects, explicitly

The main specification is a PPML gravity equation with partner fixed effects and month-year fixed effects:

$$\mathbb{E}[X_{i,t} | \mathbf{x}_{i,t}] = \exp(\alpha_i + \gamma_t + \beta \ln(1 + \text{ETR}_{i,t})), \quad (3)$$

where $X_{i,t}$ is US imports from partner i in month t , α_i is the partner fixed effect, γ_t is the month-year fixed effect, and β is the coefficient on the realized effective tariff. The mass and resistance terms from equation (1) are not written separately because the fixed effects take them in. The partner fixed effect absorbs the things about a partner that do not change during the panel: distance, language, contiguity, colonial ties, baseline FTA status, and outward resistance. The month-year fixed effect absorbs shocks that hit all partners in the same month: world income, US demand, the dollar, oil prices, freight conditions, and other common shocks.

So the regression is not comparing China to Mexico in a simple cross-section and calling the difference a tariff effect. Each partner has its own baseline. Each month also has its own baseline. What is left is the change in a partner's imports relative to its own normal level, after removing whatever that month did to everyone. The coefficient β is identified from the part of $\ln(1 + \text{ETR}_{i,t})$ that still varies after these two baselines are removed.

The partner fixed effect α_i is doing a lot of work. Distance from a partner to the United States does not change from January 2020 to January 2026. Neither does official language, contiguity, colonial history, or most persistent supply-chain relationships. Since all of these are constant inside a partner, the partner dummy absorbs them, and it also takes in the unobserved parts of the relationship, such as port infrastructure, corporate networks, and long-standing bilateral frictions.

The month-year fixed effect γ_t does the same thing for time. If the dollar is stronger in April 2025, it is stronger against all partners in that month. If oil prices, shipping costs, or US demand move in one month, that movement also affects the whole importing environment. The month dummy absorbs these common shocks. The tariff coefficient therefore comes from differences across partners inside the same time period, after each partner's own level has been removed.

3.5 The difference-in-differences design

The April 2 cutoff divides the panel into the pre-shock period, from January 2020 through March 2025, and the post-shock period, from April 2025 through January 2026. Treatment intensity is not just treated versus untreated. It is the size of the realized ETR change, which differs across partners. Some partners barely moved, especially where USMCA exemptions or the 90-day pause kept rates close to baseline. China moved much more. These low-exposure partners become the comparison group inside the month fixed effect.

The headline specification is a continuous-treatment two-way fixed effects DiD:

$$\ln X_{c,t} = \beta (\tau_c \times \mathbf{1}\{t \geq \text{Apr 2025}\}) + \mu_c + \lambda_t + u_{c,t}, \quad (4)$$

where τ_c is partner c 's treatment intensity, measured as the change in realized ETR at the April 2 cutoff, and $\mathbf{1}\{t \geq \text{Apr 2025}\}$ marks the post-shock period. The partner fixed effect μ_c removes the partner baseline, and the month-year fixed effect λ_t removes common monthly shocks. The coefficient β tells us the average post-shock change in log trade for one more percentage point of tariff exposure.

The idea is related to Boehm, Levchenko, and Pandalai-Nayar (2022), who estimate a path of horizon-specific elasticities after tariff shocks. My data do not yet run long enough to estimate a full horizon-by-horizon path with reliable standard errors, so I report the average post-shock effect over the ten months available. Boehm et al. also work at the product level. This paper uses country-month imports because the IMF IMTS outcome is country-month total trade. That loses some within-country product detail, which is why Section 5.5 later adds the product-weighted Teti tariff as a cross-check.

The diversion question uses this setup on each partner's share of total US merchandise imports, because the issue there is not only whether trade fell, but whether the United States changed which foreign supplier it used. Let $s_{c,t}$ be

partner c 's share in month t . The market-share DiD is

$$s_{c,t} = \beta (\tau_c \times \mathbf{1}\{t \geq \text{Apr 2025}\}) + \mu_c + \lambda_t + \varepsilon_{c,t}, \quad (5)$$

where τ_c is the September 2025 product-weighted applied tariff, constructed in Section 3.7. In this version, β measures how much a partner's US import share changes after the shock for one more percentage point of tariff exposure. Both specifications cluster standard errors by partner, so the errors are allowed to be correlated within a partner over time.

The identifying assumption is parallel trends. Without the April 2 shock, high-tariff and low-tariff partners should have followed similar paths in trade and market share. Goodman-Bacon (2021) discusses this problem in staggered DiD settings. Here the date is not staggered, because every partner shares the same April cutoff, but treatment intensity differs. So the relevant question is simpler: were the high-exposure and low-exposure partners already moving apart before April 2025? The pre-trend test below is designed to answer that.

3.6 Threats to identification

Anticipation. Importers may have moved shipments forward before April 2. If that happened, pre-shock trade would look too high and the later fall would look too large. The timeline makes this less likely, because the full schedule was not known far in advance and the gap between announcement and customs entry was only three days. That is too short for cargo ships from Asia to suddenly arrive at US ports. The in-transit waiver also reduced the reason to rush, since goods already loaded before April 5 were exempt. Pattararangrong and Suwanprasert (2025) find front-loading in Thai exports, but mostly after the announcement and in agricultural goods with short lead times. In the data, the formal pre-trend test gives $\hat{\beta}_{\text{pre}} = -0.0011$ with $p = 0.64$. So high-tariff and low-tariff partners were not already moving apart before April 2025.

Retaliation. Retaliation is a real part of the welfare story. China's tariffs on US exports matter, especially for soybeans, energy, and agricultural goods, but they are not the dependent variable here. Furceri et al. (2021) show that retaliation is often chosen for political salience rather than pure economic efficiency. That belongs to the broader story, but the import elasticity estimated here is based on US imports from partner i and the tariff the United States charges on those imports. Chinese retaliation operates on the other side of the trade flow.

Within-month common shocks. A partner-specific US demand shock could also happen at the same time as the tariff. Month fixed effects remove shocks that hit all partners, but they cannot remove a hidden shock that affects only one partner's goods. This does not break the design by itself, because the April 2 tariff formula was based on the bilateral trade-deficit ratio and not on a forecast of partner-specific US demand. The August 7 revision is less clean, because policy may already have been responding to early outcomes by then. Section 6 returns to that limitation.

3.7 Measurement of reshuffling, rerouting, and country-specific exposure

The market-share reallocation index measures how much US import share moved across partners. It is the O. D. Duncan and B. Duncan (1955) index of dissimilarity applied to US import shares. For partner c with average monthly trade M_c in a window, the share is $s_c = M_c / \sum_k M_k$, and

$$R = \frac{1}{2} \sum_c |s_{c,\text{post}} - s_{c,\text{pre}}| \quad (6)$$

The value of R can be read as the fraction of total US import share that changed suppliers between the pre- and post-shock windows. A quarterly version, R_t , compares adjacent quarters and is used in the time-series figure in Section 5.2.

The product-overlap share is used to ask a more specific rerouting question. It follows the idea of Grubel and

Lloyd (1971) intra-industry overlap, but here the focus is on products China used to dominate. For each HS6 product k , China's pre-shock share of US imports across the top 20 partners is θ_k . I define China-dominated products as $\mathcal{K}^* = \{k : \theta_k > 0.30\}$, which contains 1,336 products. For each partner c ,

$$o_c = \frac{\sum_{k \in \mathcal{K}^*} M_k^c}{\sum_k M_k^c} \quad (7)$$

This gives the fraction of partner c 's pre-shock US-bound exports that were in products China dominated. If a partner has high o_c and also gains US market share after the shock, that is the product-level sign of rerouting.

The country-specific applied tariff is constructed from the Teti Global Tariff Database (Teti 2025). This database records the applied US tariff on each HS6 product on each date it changed. For partner c and month t ,

$$\tau_{c,t} = \frac{\sum_k w_k^c \tau_{c,t}^k}{\sum_k w_k^c}, \quad w_k^c = M_k^{c,\text{pre}}, \quad (8)$$

where w_k^c is partner c 's pre-shock import value in HS6 product k , held fixed, and $\tau_{c,t}^k$ is the applied tariff recorded by Teti for that partner-product through the end of month t . I keep the weights fixed at the pre-shock product mix so the tariff measure does not change just because firms changed what they imported after the shock. Since $\tau_{c,t}$ varies across partners and months, it can be used as the alternative right-hand-side variable in the robustness regression of Section 5.5.

4. Data and Variables

4.1 *US bilateral trade flows*

The outcome variable is simple in idea: how much the United States imported from each partner in each month, and for that I use the IMF Direction of Trade Statistics (International Monetary Fund 2025a). IMTS reports monthly CIF (cost, insurance, freight) import values in US dollars for 235 partner countries, with the United States as the reporting country, and I download the series through the IMF SDMX 2.1 REST API with the `sdmx1` Python library, using indicator `TMG_CIF_USD`, which is total merchandise goods, CIF, in US dollars. The raw download gives 72,622 observations from January 2020 through January 2026, or 73 monthly periods, and after the missing-value check and date normalization the cleaned file contains 72,621 observations. When I keep only US merchandise imports, the panel used for the regressions and diversion analysis has 16,563 partner-month observations across 229 partner countries.

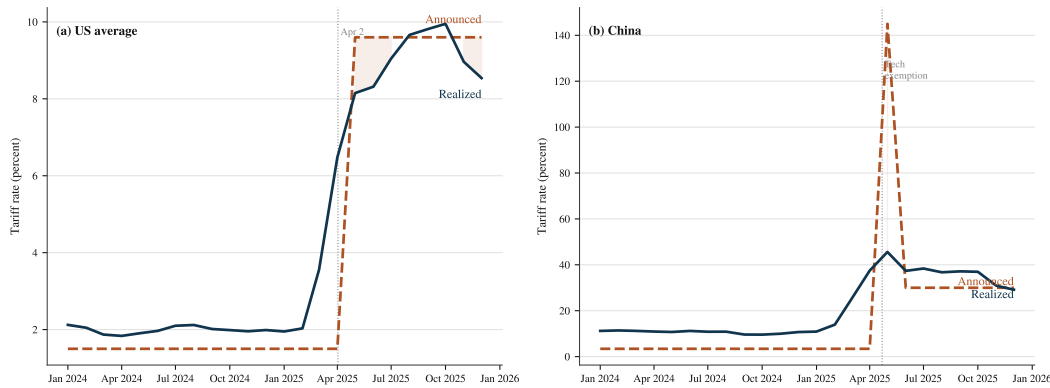
The coverage is not perfectly balanced, because small and low-income partners are more likely to have missing months in IMTS, but that does not mean the panel has to be forced into a balanced shape. PPML can keep zero trade flows in the model, while partner-months that are genuinely missing from IMTS simply stay absent from the panel. I use IMTS rather than US Census bilateral data for the outcome because the IMF source gives a wider and more consistent country coverage, while the Census bilateral tables cover around 180 trading partners.

4.2 *Realized effective tariff rate*

The tariff variable is built from what customs actually collected, because the legal rate announced by the government is not always the rate that enters a firm's cost calculation. The realized effective tariff rate (ETR) uses two monthly country-level series from the US Census Bureau (U.S. Census Bureau 2025): general import value (`GEN_VAL_M0`) and calculated customs duties (`CAL_DUT_M0`). The ETR is:

$$ETR_{i,t} = \frac{CAL_DUT_MO_{i,t}}{GEN_VAL_MO_{i,t}} \quad (9)$$

The idea behind the ratio is direct: duties collected are divided by the value of imports, so the variable moves with the burden importers actually paid rather than with the statutory schedule alone. The denominator is general import value rather than consumption import value, which makes it a good proxy for the realized burden but not an exact duty-to-dutiable-value ratio, since the value base and the duty series are not perfectly identical. Because the Census API allows only about one request every 1.5 seconds, collecting the 72 monthly pulls takes roughly eight minutes of sequential calls, and the resulting file covers 17,838 country-month observations.

Fig. II. Announced vs. Realized Effective Tariff Rate, 2025

Notes: Announced rate (dashed line) is the Yale Budget Lab weighted average statutory rate; realized ETR (solid line) is duties collected divided by general import value from US Census Bureau administrative data. The shaded region marks the gap between the two series and is driven mainly by sector exemptions (electronics, pharmaceuticals, energy products) and USMCA treatment. Source: (Yale Budget Lab 2025b); (U.S. Census Bureau 2025).

The gap in Figure II is the customs version of the policy complexity described above. The Yale Budget Lab series (Yale Budget Lab 2025b) follows the announced statutory rate, while the Census ETR follows what was actually collected after carve-outs, agreement treatment, and in-transit rules had already worked through the border. The gap is largest for China, where the headline 145 percent nominal rate overstates the realized burden substantially once the exempted and differently treated flows are removed.

ETR values above 200 percent in the raw data arise from data quality issues in the Census join, not from genuine tariff rates. These are flagged and capped at 2.0 in the processed dataset.

4.3 Gravity controls

The gravity controls are the variables that explain why some partners sell more to the United States even before the tariff enters the story. From CEPII GeoDist (CEPII 2011) I use population-weighted bilateral distance, contiguity, common official language, and colonial ties, all of which are fixed over the sample period. From CEPII Gravity Version 202211 (CEPII 2022) I use FTA/WTO membership, EU membership, and WTO membership, and because that file ends in 2020 I carry the 2020 values forward for 2021–2025. That assumption is reasonable for this short panel, since no major US bilateral agreement in the sample changed during these months, but it is still a limitation and Section 6 returns to it.

The GDP mass terms come from the IMF World Economic Outlook, April 2025 edition (International Monetary Fund 2025b), using nominal GDP in US dollars, indicator NGDPD, with annual values matched to monthly observations by year. The OECD Quarterly National Accounts (OECD 2025) are also collected for OECD members, but those quarterly values are reserved for the general equilibrium counterfactual analysis in Part 2 of the thesis and do not enter the current DiD specification. Finally, because the tariff did not hit every product chapter in the same way, I build an HS2 exemption map that records each chapter's tariff status, applicable rate, legal authority, and USMCA caveat, using USTR Annex I and Federal Register document 90 FR 15,625 as the source base. The full map is reported in Table B.

4.4 The data pipeline

I did not want the dataset to depend on manual downloads that would be hard to repeat, so all eight sources are collected, cleaned, and merged by one Python script, `pipeline.py` (Ofazzoğlu, Çağan 2026). The script runs in an isolated virtual environment with dependencies pinned in `requirements.lock`, waits between rate-limited API calls, retries

failed requests, parses unusual file formats, extracts compressed archives, and writes cached downloads to disk with a SHA-256 manifest. A full first run takes 30 to 45 minutes, while later runs take seconds because the cached files are reused, and the output is four analysis-ready processed files plus six publication-quality figures. Appendix A documents the pipeline in full.

The reason for doing this is auditability. Every processed data point should lead back to a specific API call or file download with a documented URL and access date, and the code also checks that the formula used for the ETR in the pipeline matches equation (9) in the text. If the formula in the code and the formula in the paper ever separate, the verification step flags it.

Table I. Data Accounting: From Raw Download to Analysis Samples

Stage	Unit	Countries	Months	Observations
Raw IMF IMTS download	reporter-partner-indicator-month	235	73	72,622
Cleaned import panel	partner-month	235	73	72,621
US-imports analysis sample	partner-month	229	73	16,563
Census realized-ETR panel	partner-month	varies	73	17,838
Top-40 regression sample	partner-month	40	73	2,920

Notes: The raw IMF IMTS download mixes US imports, US exports, and China exports across several reporter-indicator series. Restricting to US merchandise imports (indicator TMG_CIF_USD) yields the analysis sample. The Census realized-ETR panel is collected separately. The top-40 regression sample is the balanced panel of the 40 largest US trading partners over all 73 months. Source: author’s pipeline (Ofllazoğlu, Çağan 2026).

Table I traces the raw IMF download into the samples used later in the paper, so the observation counts in the empirical sections can be checked against one place instead of being repeated each time.

4.5 Policy periods and summary statistics

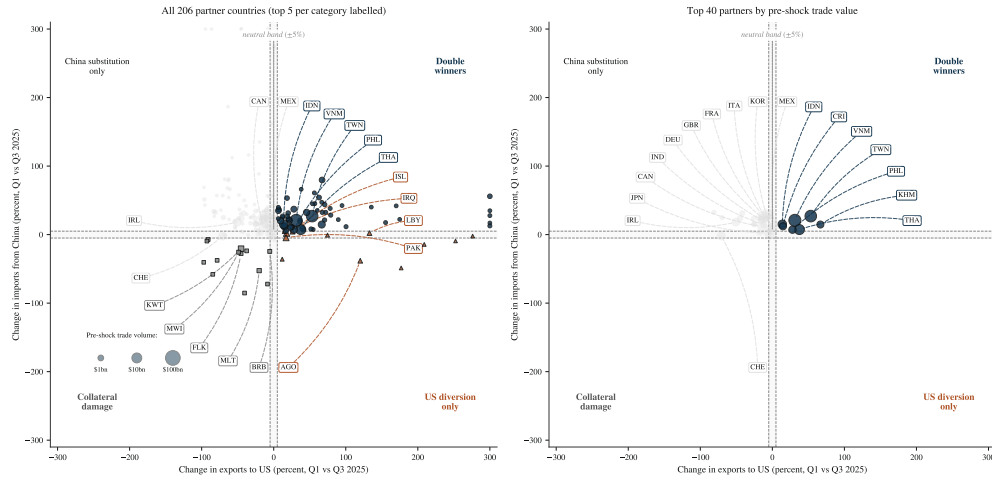
For the event-study part, each observation receives a policy-period label based on its date. The *pre-shock* period runs from January 2020 through April 1, 2025 and gives a 63-month baseline, the *shock window* runs from April 2 through July 1, 2025 and matches the announced 90-day pause, and the *post-adjustment* period begins on July 2, 2025. Table A reports the summary statistics for the full panel and for those periods separately: trade flows are right-skewed, the ETR rises sharply inside the shock window, and the panel covers 235 partners across 73 months. As a final check before the shock enters, a simple 2024 gravity diagnostic regressing log US imports on log partner GDP for the 165 partners with matched IMF WEO data gives an OLS slope of 1.27 with $R^2 = 0.69$, which is the gravity pattern already present in the data before Liberation Day.

5. Results

5.1 Descriptive findings

Before the regressions, Figure II already shows why the announced tariff cannot be treated as the burden firms actually faced. The Yale Budget Lab series gives the statutory rate announced by policy, while the Census series gives the realized ETR collected at customs, and the two begin to separate clearly after April 2. From 2020 through 2024, the average realized ETR across US trading partners stayed between 1.3 and 1.5 percent, but in 2025 it rose to 5.7 percent, more than four times its earlier level. The largest country-month realized ETR reached 70.4 percent, which is already very high, but still far below the 145 percent China headline rate.[†] The reason is the one described in the policy section: large parts of trade moved through exemptions, special treatment, or timing rules before customs collected the duty. The burden also differs sharply across partners, since the latest month ranges from zero to more than 50 percent at the country level, with the highest rates concentrated among the partners most exposed to the new schedule while low-exposure groups stay near zero.

Fig. III. Trade Diversion: Export and Import Changes, Q1 vs Q3 2025



Notes: Each axis is an ordinary percentage change between two quarterly totals: $x = (M_{Q3}^c - M_{Q1}^c) / M_{Q1}^c \times 100$ where M^c is partner c 's US-bound exports, and the y axis is the same construction for imports from China. Values of -100 correspond to a partner whose US-bound exports went to zero between Q1 and Q3 2025; values below -100 are not possible (they would require negative trade) and any point on the figure that visually sits below -100 is the effect of the ± 300 winsorisation cap applied to a few extreme positive outliers. The shaded square marks the ± 5 percent neutral band. Marker shape encodes category (circles for double winners, triangles for US-diversion recipients, squares for collateral damage, small dots for neutral). Bubble area is proportional to Q1 2025 US import value. Labels printed on a cream background are IMF IMTS aggregate groupings; the rest are individual partners by ISO 3 code. Source: IMF IMTS and author's calculations.

Once the customs burden and the policy headline have been separated, the next question is where the trade moved. Figure III compares each partner's change in exports to the United States with its change in imports from China, using Q1 2025 as the pre-shock quarter and Q3 2025 as the post-shock quarter. A partner is counted as a *double winner* when both axes rise by more than 5 percent, as a *US-diversion recipient* when US-bound exports rise while the China-imports axis is flat or missing, as *collateral damage* when both fall by more than 5 percent, and as *neutral* otherwise. With that neutral band, the 228 partners divide into 60 double winners, 13 US-diversion recipients, 15 collateral-damage cases,

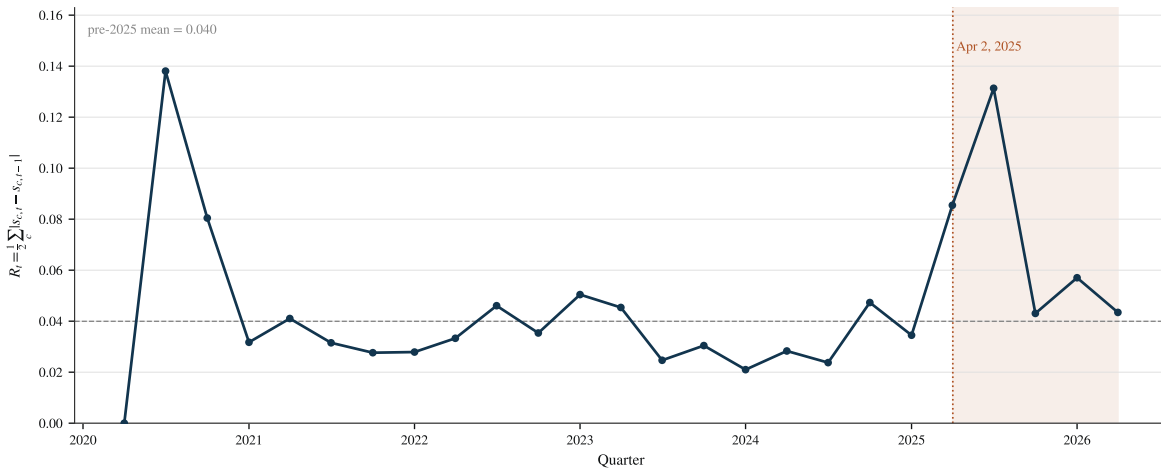
[†] The ETR is computed as CAL_DUT_MO / GEN_VAL_MO using US Census Bureau administrative records. Values above 200 percent in the raw data arise from data quality issues in the country code merge and are capped at 2.0 in the processed file.

and 140 neutrals, while Table D shows that the broad result is not created by the exact cutoff. Among partners that cross at least one threshold, the largest group is the one that both sells more to the United States and buys more from China, which is exactly the joint movement one would expect if some trade is being routed through third countries. The country-level data cannot prove that the same physical goods moved through a different route, because they only show declared partner origin and aggregate flows, so Section 5.3 adds the product-level test that is needed before calling the pattern rerouting.

5.2 Did US import sourcing actually reshuffle?

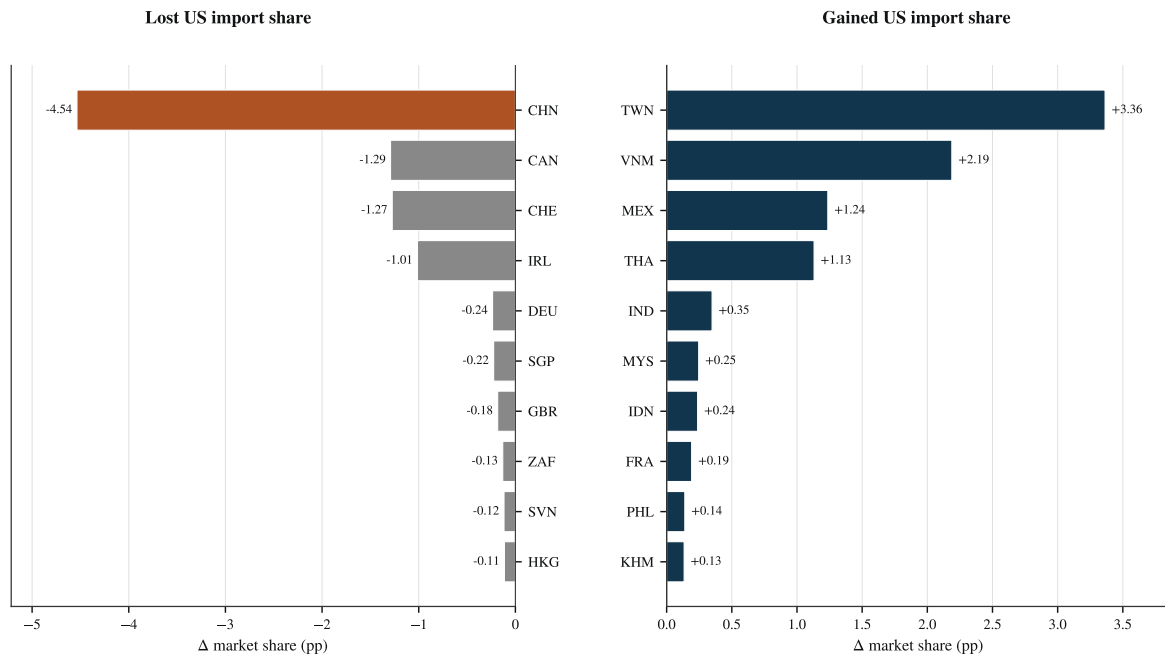
The diversion scatter tells the direction of adjustment, but categories by themselves do not tell how much of the US import market actually changed hands. To put a size on the reshuffle, I compare each partner's share of total US merchandise imports in the pre-shock window, January 2024 to March 2025, with its share in the post-shock window, April 2025 to January 2026, where the share of partner c is $s_c = M_c / \sum_k M_k$ and M_c is average monthly US imports from that partner.

Fig. IV. Quarterly Market-Share Reallocation Index, 2020–2026



Notes: The quarterly reallocation index $R_t = \frac{1}{2} \sum_c |s_{c,t} - s_{c,t-1}|$ measures the share of US import market share that moved across partners relative to the previous quarter. The dashed line marks the pre-2025 mean (0.04); the vertical dotted line marks Liberation Day (April 2, 2025) and the shaded region the post-shock window. Source: IMF IMTS and author's calculations.

The market-share reallocation index R from equation (6) is 0.10 after the shock, meaning that roughly ten percent of US import market share moved across suppliers. That number only becomes meaningful when it is compared with normal reshuffling, so I recompute the same index for three placebo window pairs rolled back one, two, and three years. Those placebo values are 0.05, 0.04, and 0.04, which means the 2025 movement is almost twice the largest comparable pre-2025 reshuffle. The quarterly version in Figure IV tells the same story from another angle: Q2 2025 reaches $R_t = 0.13$, more than three times the pre-2025 quarterly mean of 0.04 and matched in this sample only by the COVID-onset quarter of 2020.

Fig. V. Top Gainers and Losers in US Import Market Share

Notes: Change in each partner's share of total US merchandise imports between the pre-shock window (January 2024 to March 2025) and the post-shock window (April 2025 to January 2026), in percentage points. The ten largest losers are shown on the left and the ten largest gainers on the right; China is highlighted in sienna. Source: IMF IMTS and author's calculations.

Table II. Largest Gainers and Losers in US Import Market Share

Partner	Pre share (%)	Post share (%)	Δ share (pp)	Rank (pre→post)
<i>Gainers</i>				
Taiwan	3.57	6.93	+3.36	9 → 4
Vietnam	4.18	6.37	+2.19	6 → 5
Mexico	15.12	16.35	+1.24	1 → 1
Thailand	1.92	3.05	+1.13	14 → 10
India	2.73	3.08	+0.35	11 → 9
Malaysia	1.59	1.84	+0.25	16 → 15
<i>Losers</i>				
China	12.85	8.31	-4.54	2 → 3
Canada	12.37	11.08	-1.29	3 → 2
Switzerland	2.99	1.72	-1.27	10 → 16
Ireland	3.87	2.85	-1.01	8 → 11
Germany	4.80	4.57	-0.24	4 → 6
Singapore	1.29	1.07	-0.22	17 → 19

Notes: Market share is each partner's share of total US merchandise imports, computed from average monthly import values in the pre-shock window (January 2024 to March 2025) and the post-shock window (April 2025 to January 2026). Δ share is in percentage points. Ranks are by import value within each window. The full distribution appears in Figure V. Source: IMF IMTS and author's calculations.

The reshuffle was large in market-share terms, but it was not a complete collapse of the old supplier hierarchy. When partners are ranked by import value before and after the shock, the two orderings still line up almost perfectly: the

Spearman rank correlation is 0.99, Kendall's τ is 0.92, and none of the top twenty suppliers drops out of the top twenty. The large players therefore stayed large, but their weights changed inside the group, which is exactly why Table II and Figure V are useful. China lost 4.5 percentage points of US import share, by far the largest single movement, falling from 12.9 to 8.3 percent and from second to third place. Taiwan gained the most, while the rest of the gain mostly went to Asian and North American suppliers rather than to US production, and the non-China losers were concentrated in high-value exemption-sensitive corridors. The size and shape of the reshuffle therefore fit the idea of rerouting through third countries, but the country-level evidence still cannot by itself prove that the same physical goods were rerouted, which is why the product-level evidence below is needed.

Table III. Difference-in-Differences on US Import Market Share

	(1) DiD on share	(2) Pre-trend placebo
$\tau_c \times \mathbf{1}\{t \geq \text{Apr 2025}\}$	−0.0681** (0.0299)	
$\tau_c \times (\text{pre-shock time trend})$		−0.0011 (0.0024)
Partner FE	Yes	Yes
Month FE	Yes	Yes
Sample	Top 20 partners	Top 20 partners
Window	Jan 2024–Sep 2025	Pre-shock months only
N partner-months	420	300
Within- R^2	0.32	0.18

Notes: OLS estimates of equation (5). The dependent variable $s_{c,t}$ is partner c 's monthly share of US merchandise imports in percentage points. The treatment intensity τ_c is the September 2025 product-weighted applied tariff for partner c , fixed over time. Column (1) reports the post-shock DiD coefficient: a one percentage point higher treatment intensity is associated with a 0.068 percentage point lower monthly market share in the post-shock window, significant at the 5 percent level. Column (2) is the pre-trend placebo on the pre-shock months only, with the post-shock indicator replaced by a linear time trend; the interaction is statistically and economically zero, supporting parallel trends. Standard errors clustered by partner in parentheses. Significance: * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

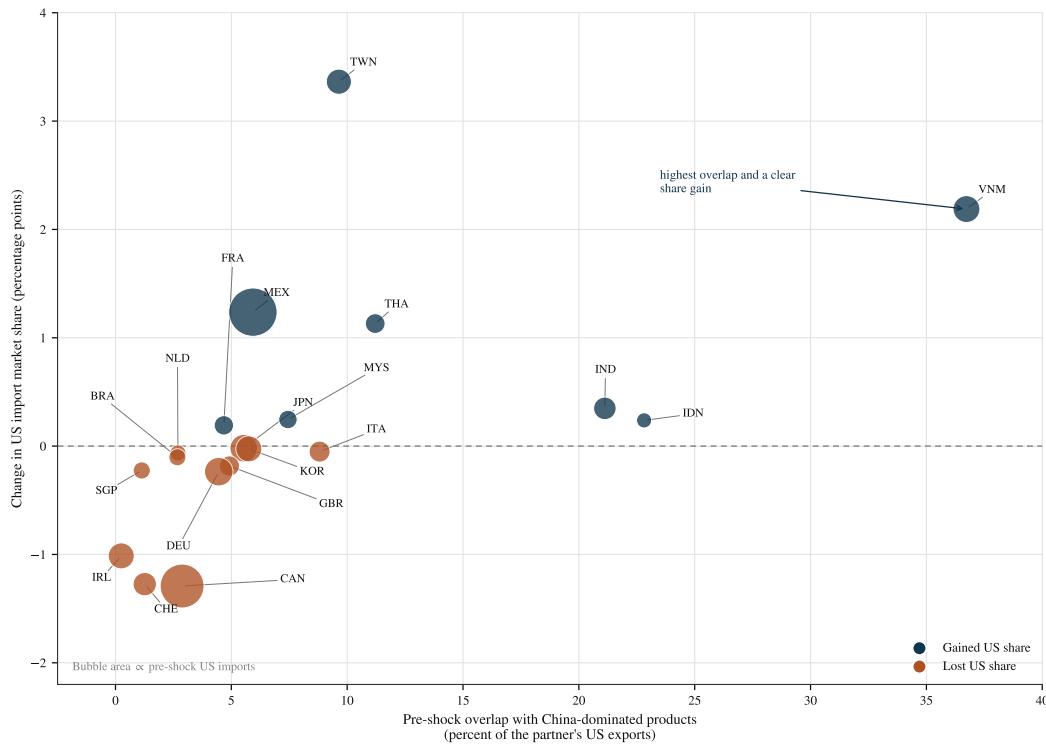
The raw before-and-after shares are not enough on their own, because a partner could have been losing US market share before the tariff even arrived, so I estimate the market-share DiD from equation (5). In Table III, a one percentage point higher September 2025 tariff is associated with a 0.068 percentage point lower monthly market share after the shock. For China, whose product-weighted tariff reached 59.3 percent, that coefficient predicts a share fall of about −4.0 points, which is close to the −4.5 points observed in Table II. The pre-trend placebo in column (2) is close to zero, so high-tariff and low-tariff partners were not already drifting apart before April 2025 in a way that would explain the result.

5.3 The product-level signature of rerouting

Customs data do not observe rerouting directly, because the customs record reports the declared country of origin rather than the full upstream supply chain that produced the good. If a product uses Chinese intermediates, moves through another country for final assembly, and then enters the United States under that partner's origin, the country-level panel sees the final partner, not the whole path. Therefore I infer rerouting only when three pieces line up together: China

loses US market share, another partner gains both US exports and Chinese imports, and that partner's gains are in the same HS6 product lines that China previously dominated. By *rerouting* I mean this substitution of one foreign supplier for another in the same kind of good, because if the United States used to buy a product category from China and then buys that same category from a different supplier after the tariff, the source changed even if production did not come home.

Fig. VI. Product-Level Overlap with China and the Change in US Market Share



Notes: Each bubble is one of the 19 largest US partners other than China. The horizontal axis is the share of the partner's pre-shock (through March 2025) US exports that fell in the 1,336 HS6 products where China held more than 30 percent of US imports; the vertical axis is the change in the partner's share of total US imports between the pre-shock and post-shock windows. Bubble area is proportional to pre-shock US imports. Navy marks partners that gained US share, sienna those that lost it; China is the reference category and is omitted. Vietnam pairs the highest product overlap with a clear share gain, the product-level signature of rerouting, while Taiwan gained the most share on low overlap because its strength sits in its own semiconductors. Source: US Census HS6 import records and author's calculations.

The market-share reshuffle and the diversion classification are therefore only the first part of the test, because a gain in US share could be rerouting or it could simply be a partner selling more of goods it already specialized in. To separate those two possibilities, I move to six-digit product records for the top 20 partners and identify the 1,336 HS6 products in which China held more than 30 percent of US imports before the shock. For each partner I then calculate what fraction of its own pre-shock US exports fell inside that China-dominated set, and that fraction is the partner's *product overlap* with China. A partner with high overlap sells the same kinds of goods China used to sell to the United States, so if that partner also gains US market share after the shock, the gain carries a product-level fingerprint of replacement that the country-level data cannot show. Appendix C documents the procedure and `outputs/product_exposure.csv` holds the full result, while Figure VI plots overlap against the change in US market share.

The overlap splits the gainers in a way that makes the rerouting story more specific. Vietnam, the second-largest

gainer at plus 2.2 percentage points of US share, had 37 percent of its pre-shock US exports in China-dominated products, by far the highest overlap in the group, so more than a third of what it was already sending to the United States sat in categories where China had been the leading supplier. When Vietnam then gains US share after the shock, the gain falls exactly where rerouting should appear. Other Asian partners near China's supply chains show the same pattern at lower intensity, while Mexico's lower overlap means its gain is more about its own product base than about replacing China. Taiwan shows the other side of the point: it gained the most share, plus 3.4 points, but overlapped China in only 10 percent of products, because its gains sit in semiconductors and electronics where Taiwan was already strong. The largest non-China losers also point away from a simple China-substitution story, since their overlap with China was very low and their losses run more through high-value exemption-sensitive sectors than through direct competition with Chinese goods.

The result is that rerouting appears real but selective. It is strongest where the partner already made the same kinds of goods China exported, and it is weak or absent where the gain came from the partner's own specialized position. That fits the modest, reroute-existing-flows diversion that Fajgelbaum and Khandelwal (2022) document for the 2018 trade war, and it sharpens the country-level finding by showing which gains carry China's product signature and which do not.

5.4 *Regression estimates*

Table IV. Main PPML Gravity Regression: Bilateral US Imports, 2020–2025

	(1) PPML	(2) OLS on log trade
$\ln(1 + \text{ETR}_{c,t})$	-1.675^{**} (0.787)	-0.727 (1.008)
Intercept	21.21^{***} (0.016)	21.22^{***} (0.030)
Country fixed effects	Yes	Yes
Month-year fixed effects	Yes	Yes
N observations	2,920	2,920
Pseudo R^2 / R^2	0.994	0.979

Notes: Sample is the top 40 US trading partners (by total merchandise import value) over the 72 months from January 2020 through December 2025 with non-missing country-month realized ETR. The realized ETR varies by partner and by month, constructed from US Census duties and import values per equation (9) and merged onto the bilateral panel through the Census name to ISO3 crosswalk built in Section 4. Column (1) reports Poisson Pseudo Maximum Likelihood estimates using `statsmodels.GLM` with Poisson family and log link, the Santos Silva and Teneyro (2006) estimator. Column (2) reports OLS on the log of positive trade flows for comparison; the smaller absolute coefficient there reflects the heteroskedasticity bias Santos Silva and Teneyro document. Standard errors in parentheses are clustered by partner country. Country fixed effects absorb every time-invariant partner characteristic (distance, language, contiguity, colonial ties, baseline trade level, persistent unobserved bilateral frictions). Month-year fixed effects absorb every shock that hits all partners in a month (the dollar, US business cycle, global shipping costs, world commodity prices). The coefficient β approximates a tariff semi-elasticity because $\ln(1 + \text{ETR}) \approx \text{ETR}$ for small rates. The full-sample robustness check on 228 partners with country-month ETR yields $\hat{\beta} = -1.59$ ($p = 0.037$), reported in Section 5.5. Significance: $^{\dagger} p < 0.15$, $* p < 0.10$, $** p < 0.05$, $*** p < 0.01$. Stars use partner-clustered standard errors.

The gravity regression asks whether the tariff burden that appears in the customs data is also visible in trade flows after partner baselines and common monthly shocks are removed. Table IV uses the top 40 US trading partners, restricted to US merchandise imports, giving 2,920 country-month observations with non-missing realized ETR. The specification follows equation (3): PPML with the realized ETR as the continuous treatment, partner fixed effects, and month-year fixed effects, where the ETR is merged to the bilateral panel through the Census-name to ISO3 crosswalk described in Section 4. The model explains 99.4 percent of the variation in bilateral trade.[†]

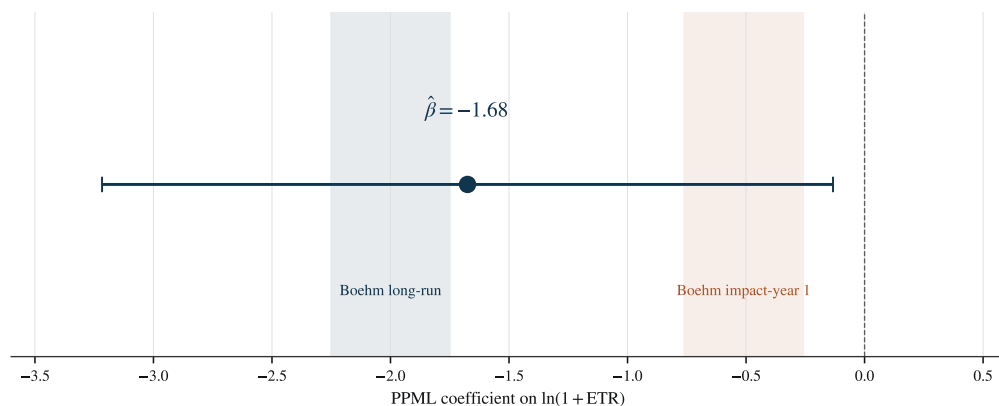
The coefficient on $\ln(1 + \text{ETR})$ is -1.68 , with a partner-clustered standard error of 0.79 and $p = 0.033$, so a one percentage point increase in a partner's realized effective tariff is associated with about a 1.7 percent decline in US imports from that partner, after holding fixed the partner's usual level and the shocks common to that month. The estimate sits just below the long-run band of -1.75 to -2.25 in Boehm, Levchenko, and Pandalai-Nayar (2022), which is large for such a short post-shock window, but the realized rate is also closer to what firms actually paid at customs than a statutory schedule alone. When this result is read together with the Teti applied-rate regression in Section 5.5,

[†] Pseudo R-squared is one minus the ratio of the model deviance to the null deviance, the PPML analogue of OLS R-squared.

which gives -1.44 , the two estimates tell the same story from different sides of the tariff: legal exposure and customs incidence both point to lower imports.

Column (2) shows why PPML is the preferred estimator here. The log-OLS coefficient is smaller in absolute value, -0.73 , with a larger standard error and no conventional statistical significance, which is exactly the kind of movement Santos Silva and Tenreyro (2006) warn about when heteroskedastic trade data are forced into a log-linear model. The full-sample check on 228 partners with country-month ETR gives $\hat{\beta} = -1.59$ with $p = 0.037$, so the negative elasticity is not created by looking only at the largest partners.

Fig. VII. PPML Coefficient on $\ln(1 + \text{ETR})$ vs the Boehm et al. Band



Notes: The main PPML estimate of the elasticity of bilateral US imports with respect to one plus the realized effective tariff rate, with its 95 percent partner-clustered confidence interval ($\hat{\beta} \pm 1.96 \cdot \text{SE}$), placed next to the Boehm, Levchenko, and Pandalai-Nayar (2022) short and long-run elasticity bands. The estimate is unusually large for the short window available here, which is consistent with the realized rate capturing more of the actual tariff burden at the border than the applied rate used in Boehm et al. Source: author's PPML regression on the IMF IMTS bilateral panel with country-month ETR.

Figure VII places the main coefficient and its 95 percent confidence interval next to the Boehm, Levchenko, and Pandalai-Nayar (2022) short- and long-run bands, which makes the size comparison visible rather than leaving it only inside the paragraph.

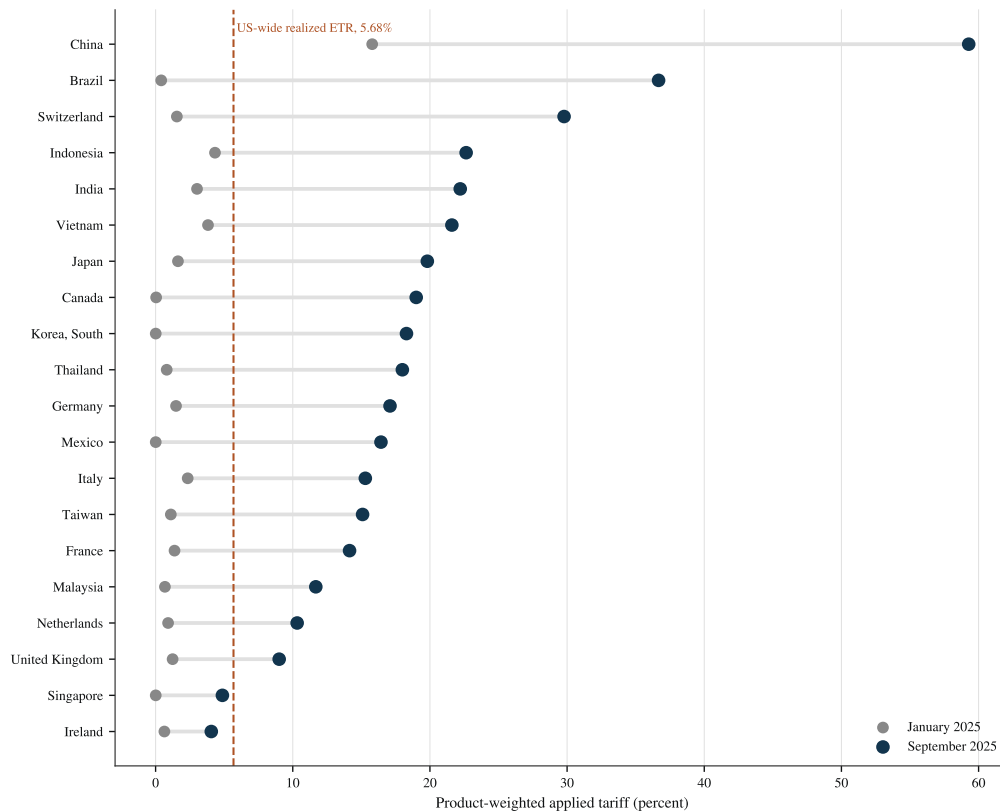
5.5 Country-specific applied tariff: a second independent measure

The main specification uses the realized ETR, which is the duty burden observed at customs, but it is still useful to check whether the same result appears when the tariff is measured from the legal schedule before customs timing and exemptions are netted out. For that I build a country-specific applied tariff from the Teti Global Tariff Database (Teti 2025), which records the US applied rate for each partner-product combination on each date the rate changed between 2018 and 2025. I weight those HS6 rates by each partner's own pre-shock product mix and hold the weights fixed, so the measure changes when the tariff schedule changes rather than when firms change what they import. The result is a second tariff measure that varies across both countries and months, and I use it to re-estimate the gravity regression.

Table V. Country-Specific Product-Weighted Applied Tariff (percent)

Partner	Jan 2025	Apr 1, 2025	Sep 2025
China	15.8	36.8	59.3
Brazil	0.4	2.6	36.7
Switzerland	1.6	1.6	29.8
Vietnam	3.8	4.2	21.6
India	3.0	3.9	22.2
Japan	1.6	2.2	19.8
Mexico	0.0	12.7	16.4
Germany	1.5	2.3	17.1
Taiwan	1.1	1.8	15.1
United Kingdom	1.2	1.8	9.0
Ireland	0.6	0.7	4.1

Notes: Each partner's applied tariff from Teti (2025) weighted by that partner's pre-shock (through March 2025) six-digit product mix, weights held fixed. The US-wide realized ETR for 2025 averaged 5.68 percent. Source: Teti Global Tariff Database, US Census product detail, and author's calculations.

Fig. VIII. Country-Specific Product-Weighted Applied Tariff, January versus September 2025

Notes: Each partner's Teti applied tariff weighted by its pre-shock six-digit product mix, weights held fixed. Grey marks January 2025, navy marks September 2025, and the dashed line is the US-wide realized ETR of 5.68 percent. The spread runs from 4.1 percent for Ireland to 59.3 percent for China. Source: Teti Global Tariff Database, US Census product detail, and author's calculations.

The cross-country spread in the applied measure is wide. In September 2025 the product-weighted rate runs from 4.1 percent for Ireland to 59.3 percent for China, against the US-wide realized average of 5.68 percent. China's number is still far below the 145 percent headline, because its product mix passes through the same carve-outs discussed

earlier, but it remains the highest country-specific exposure by a large margin. Table V lists representative partners, and Figure VIII plots all twenty.

When the gravity regression is re-run with this applied tariff instead of the realized ETR, using the top 20 partners from January 2024 to September 2025, the PPML coefficient on $\ln(1 + \text{tariff})$ is -1.44 with a heteroskedasticity-robust standard error of 0.495 and $p = 0.004$. The coefficient is negative, significant, and inside the -0.76 to -2.25 short-run band of Boehm, Levchenko, and Pandalai-Nayar (2022). Since the Census realized-rate result is -1.68 and the Teti applied-rate result is -1.44 , the two measures differ in level but not in direction, which means the negative elasticity does not depend on whether the tariff is measured from the legal schedule or from customs collections.

The applied-tariff check still has limits, because the Teti weights are fixed at the pre-shock product mix, which rules out within-product substitution, and because the estimate is partial equilibrium and short run. However, the earlier worry about relying on the top 20 partners is answered by the full-sample realized-rate result in Section 5.4: on 228 partners with country-month ETR, $\hat{\beta} = -1.59$ with $p = 0.037$. The applied-tariff measure could be extended to all partners with a complete country-code crosswalk, but for this paper the realized rate already covers the full sample and gives the same qualitative answer.

5.6 Robustness of the country-specific elasticity

The applied and realized tariffs correlate at 0.90 across the panel, which is why the result does not hinge on one side of the customs counter. Re-running the same exercise on the realized collected rate gives -2.79 with a heteroskedasticity-robust standard error of 0.294, and the full sample of 155 reporting partners gives -2.67 with clustered standard error 0.471 and $p < 0.001$. Partner-clustered standard errors on the headline -1.44 are 0.150, tighter than the HC robust 0.495, while the pre-trend test on treatment intensity interacted with the pre-shock time trend is essentially zero at $p = 0.51$. The same intensity interacted with the post-shock indicator is large, negative, and significant at $p < 0.001$, and Larch (2026) confirms that no US bilateral trade agreement among these partners changed in 2024–2025, so the held-constant CEPII agreement flags are not doing the work.

6. Discussion and Limitations

6.1 *Short-term versus long-term horizon*

Time is the clearest limit here. The post-shock data in this paper run only from April 2025 to January 2026, so the estimates are short-run whether one wants them to be or not. Boehm, Levchenko, and Pandalai-Nayar (2022) show that tariff elasticities can take years to reach their long-run value. So the result should not be read as the final trade response. If anything, it is closer to an early lower bound on what the policy may eventually do.

6.2 *Policy uncertainty*

Uncertainty is another limit. The 90-day pause did not create a stable policy environment; it created a temporary one, and then that temporary one was extended. Handley and Limão (2015) show that policy uncertainty can reduce trade even if the applied tariff rate itself does not move. The data here cannot tell whether importers expected the original higher rates to come back, or whether they believed the pause would keep being extended. The pre-trend test rules out the clearest anticipation problem before April 2, but it cannot separate the tariff effect from the uncertainty effect after the shock.

6.3 *Within-country sector heterogeneity*

The main customs burden is also measured at the country-month level. In practice, that means total customs duties divided by total import value. But countries do not export only one product. A country can ship exempt electronics and tariffed furniture at the same time, and the country-level average will smooth those two very different exposures together. For Japan, which ships many electronics and machinery products that were carved out, the average can hide the burden on the smaller tariffed part of its exports. For Vietnam, where apparel and furniture matter more, the same average is closer to the actual burden. This is not a flaw that destroys the estimate, but it is a reason to be careful about interpreting a country-level tariff as if every sector in that country paid the same rate.

6.4 *Sector mix*

The product-weighted tariff in Section 5.5 is meant to reduce this problem. It uses the Teti Global Tariff Database (Teti 2025) and weights each partner's tariff by its own pre-shock product mix. With that measure, the coefficient is -1.44 , less negative than the realized-rate estimate. The direction of that gap is itself informative, because it shows that the product schedule and the customs burden do not measure the same object. The cleaner version of this exercise would be a partner-product panel, where product fixed effects and partner fixed effects could sit in the same model. That is beyond the current country-month panel.

6.5 *Bias from tariff endogeneity*

Tariffs were also not randomly assigned. The administration's reciprocity formula set each partner's rate to half of its bilateral trade-deficit ratio with the United States. If this formula gave higher tariffs to countries whose exports to the United States were already falling before April 2025, then the coefficient would look too negative. So the pre-trend test has to do real work here. The interaction between eventual treatment intensity and a pre-shock time trend is -0.0011 with $p = 0.64$, which is statistically and economically zero. So the data do not support the most obvious endogenous-assignment story.

6.6 *OLS bias from heteroskedasticity*

Another possible bias would come from using the wrong estimator. Santos Silva and Tenreyro (2006) show that log-linear OLS is inconsistent when the variance of trade depends on the regressors, which is exactly what happens in

bilateral trade data. This paper uses PPML to avoid that problem. PPML estimates the conditional mean directly, keeps the dependent variable in levels, and does not require the same constant variance assumption.

6.7 *CEPII gravity controls and time coverage*

The CEPII Gravity dataset ends in 2020, so FTA membership and WTO status are held constant for 2021–2025. This is not ideal, but it is not a large problem for this paper because no major US bilateral trade agreement with a substantial trading partner was created or removed in this window. When a time-varying agreement flag is needed, the pipeline can use Mario Larch’s Larch (2026) Regional Trade Agreements Database, which extends these indicators through 2025.

6.8 *Cross-source validation of the trade figures*

Because the results rely heavily on US Census customs data, I check the trade figures against USITC DataWeb. I do this because USITC is also an official source, but it reports customs value through a separate accounting system. Across the top 20 partners in Q1 and Q3 of 2025, the mean absolute difference is 1.9 percent. The largest gap is 7.7 percent, and Mexico and Canada agree within 0.6 percent. Therefore, the trade figures used here do not seem to be an artifact of one source. The procedure is documented in Appendix C, and the partner-level gaps are saved in `outputs/usitc_census_discrepancy.csv`.

6.9 *General equilibrium effects*

General equilibrium is the limit that sits outside the country-month estimate. The estimates in this paper measure bilateral trade responses while holding the multilateral resistance terms fixed. In reality, when tariffs change, the price indices P_j and Π_i in equation (1) also move. A tariff on one partner can affect other partners through substitution, prices, and third-country routes. A full welfare counterfactual would need the steps in Yotov et al. (2016): estimate the gravity equation, construct the bilateral trade-cost matrix, and then solve the multilateral resistance system again.

It is still useful to see the order of magnitude. Using the gains-from-trade formula in Costinot and Rodríguez-Clare (2018), and taking BEA data that put US imports at roughly 8 percent of final demand, the domestic expenditure share is about 0.92. Using the realized ETR change estimated earlier gives $\Delta \ln(1 + \text{ETR}) \approx 0.042$. Applying the country-specific elasticity of -1.44 gives $-1.44 \times 0.042 \approx -0.060$, or an estimated 6 percent decline in imports, lifting the domestic share to about 0.925. With a standard long-run elasticity of 5, this gives a static welfare loss of about 0.1 percent of real income; using the short-run elasticity estimated here gives about 0.4 percent. This is only a back-of-the-envelope calculation, not a full general-equilibrium model, but it is in the same broad range as the -0.13 to $+0.10$ percent of GDP reported by Fajgelbaum and Khandelwal (2026).

7. Conclusion

The word “liberation” was the beginning of this paper because it was also the beginning of the political story told around the tariff. Berlin warned that a policy can claim to free people while still restricting them in practice. The question here was whether something similar happened in trade policy.

The results show that Liberation Day was not only a symbolic event. Tariffs rose, trade responded, and the sourcing map of the United States changed. But the same results also make the political meaning harder to defend. The tariff schedule announced to the public was not the same as the burden that passed through customs, and the movement away from China mostly looked like movement toward other foreign suppliers, not like a clear return of production to the United States.

So was Liberation Day a delusion? The answer is not that nothing happened. Something clearly happened. But if liberation meant freeing the American economy from foreign dependence and lowering the burden on American buyers, the evidence is much weaker. A large part of the adjustment looks like a change in supplier, not a change in dependence. The factory did not necessarily come home. The address on the shipping route changed.

So the word cannot be treated as just a slogan. A tariff can be described as liberation, but the customs data cannot be made to say that the American buyer was necessarily freed. The policy raised money for the government, changed who sold goods to the United States, and shifted market share across foreign partners. It did not clearly prove that the American people were better off. In that sense, the results bring the paper back to Berlin: freedom depends on what is meant by freedom. If it means fewer foreign suppliers and lower costs for American buyers, the evidence does not show that. If it means a different map of foreign suppliers, then Liberation Day did achieve something, but it was not the liberation that the word promised.

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APPENDIX A

Data Pipeline Documentation

A.1 Overview and motivation

This appendix is included because AI was part of the process, and I do not want that part to be hidden behind the final PDF as if every line had arrived in exactly the same way. The responsibility for the thesis is still mine, but if software helped with code, search, explanations, or drafting, then the reader should know where that help entered and where it stopped.

The data pipeline, `pipeline.py`, was built with help from an AI coding assistant, but the process was not that I asked for a thesis dataset and accepted whatever came back. I gave instructions function by function: which source had to be downloaded, which dates and countries had to be covered, how long the script had to wait between API calls, how country names had to be crosswalked, and what the final processed files had to look like. The assistant wrote draft code from those instructions, after which I ran the code, checked the errors, changed prompts or code when needed, and repeated the process until the pipeline returned files that could be checked against the original sources. Because of that, the script documents endpoints, retry logic, waiting times, URLs, access dates, and cached files, so the reader does not have to trust an invisible data step.

The figure script, `figures_v2.py`, was made in the same kind of back-and-forth. AI helped with technical plotting tasks such as label placement, confidence bands, the market-share mirror chart, the reallocation-index figure, and the coefficient plot, but the choices about what each figure should show, which variables mattered, how the colours should work, and what visual style fitted the thesis were mine. The same division applies to the rest of the technical work: AI helped with retry loops, fallback URLs, error handling for unavailable files, and formatting of tables, figures, references, and LaTeX environments, while I decided what should be automated and what had to be checked manually.

I also used AI-assisted search to locate data sources and papers, including the tariff files, customs data, gravity datasets, agreement data, and the main papers cited in the thesis. Finding a source, however, is not the same thing as using it correctly, so the decision to include each source and the reason it belongs in the argument are mine. For theoretical parts that were harder to learn, I sometimes asked AI to explain the idea back to me in simpler language, especially for PPML versus log-OLS, multilateral resistance, fixed effects, and tariff elasticity bands. Those explanations were a study aid, not a substitute for the original papers or for the way the estimates in this thesis are interpreted.

Some writing also passed through AI-assisted drafting, because I sometimes started from my own notes, asked for a rough paragraph, and then rewrote it until it sounded like the argument I wanted to make. I also used a humanisation tool on paragraphs that still sounded too artificial, but those outputs were edited again before entering the LaTeX source. The final wording is therefore not a raw model output; it is a revised version that I accept responsibility for.

The parts I do not treat as delegated are the research questions, the identification strategy, the decision to use PPML gravity with two-way fixed effects, the continuous-treatment DiD on partner-month shares, the pre-trend placebo, the choice to connect the literature and empirical strategy in Section 3, the interpretation of the coefficients, the policy implications, the Berlin and Liberation framing, the final figure and data-source choices, and the final section-by-section read-through. If there is a mistake in the specification, the interpretation, or the writing, it is mine.

The pipeline itself is a single Python script that collects, cleans, and merges all eight datasets used in this paper, because collecting the files by hand would make the thesis much harder to audit. Every number in the processed output should trace back to a specific API call or file download with a documented URL and access date.

The pipeline runs under Python 3.12 in an isolated virtual environment. Dependencies are pinned in `requirements.lock`, and the standard installation is shown below. No external API keys are required; all data sources are publicly accessible without authentication.

A.2 Usage

```
# One-time setup in an isolated environment.
python -m venv .venv
source .venv/bin/activate
pip install -r requirements.lock
```

```
# Full run: download, process, generate all figures and tables.
python pipeline.py run

# Download raw data only, without processing.
python pipeline.py download

# Audit that the ETR code matches its formula in this paper.
python pipeline.py verify-latex

# Run the PPML gravity regression (statsmodels GLM, Poisson family).
python pipeline.py regress
```

A.3 Data source reference

Source	Access method	Frequency	Coverage
IMF IMTS	SDMX 2.1 REST API	Monthly	2020–2025
US Census Bureau	JSON API	Monthly	2020–2025
Yale Budget Lab	XLSX download	Daily	Apr–Jul 2025
CEPII Gravity	ZIP download	Annual	1948–2020
CEPII GeoDist	ZIP download	Static	All pairs
IMF WEO	XLS download	Annual	1980–2029
OECD QNA	SDMX JSON API	Quarterly	1970–2025
Teti Global Tariff DB	Dropbox (access-restricted)	Daily	2018–2025

A.4 Pipeline architecture

Figure A shows the pipeline as a directed acyclic graph.

Sources (top layer) feed into download functions (second layer), which feed into processing functions (third layer), which produce CSV outputs and figures (bottom layer). Edges represent data dependencies. The IMF IMTS data feed into three processing steps; all five gravity control datasets feed into the event study panel.

A.5 Known fragile points

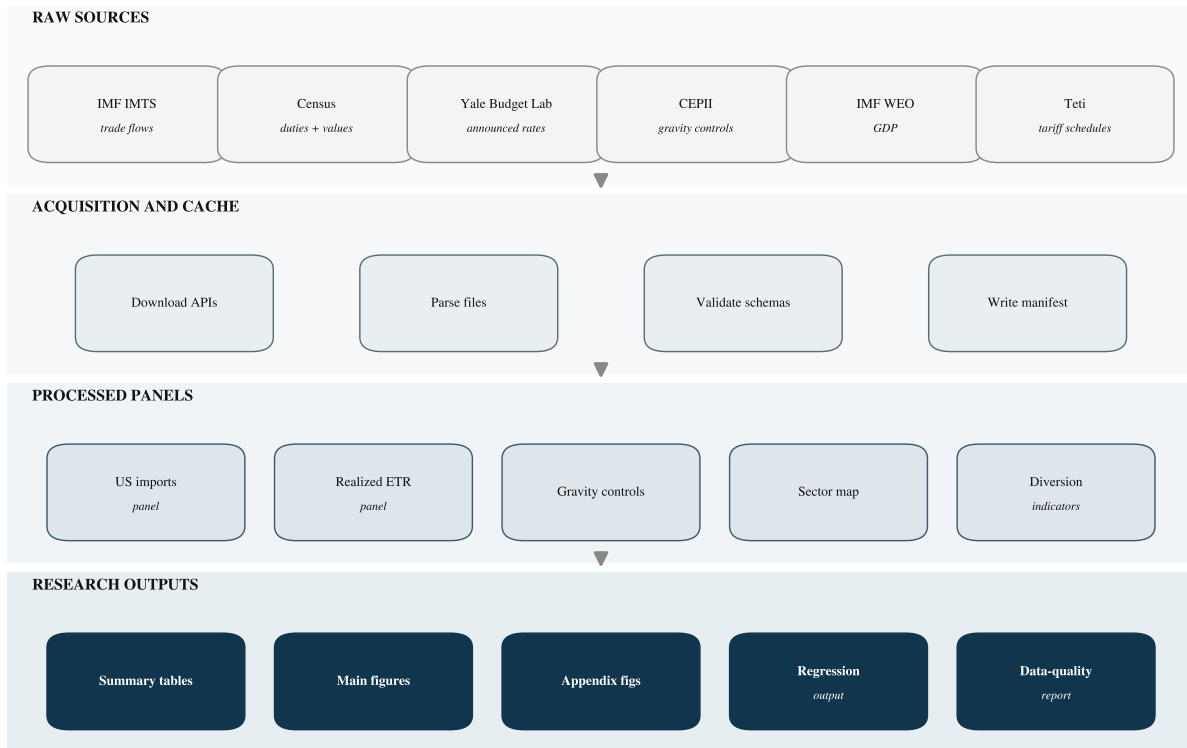
Four components of the pipeline are known to fail or behave unexpectedly under specific conditions.

The Yale Budget Lab URL changes monthly. The pipeline tries three candidate URLs constructed from recent dates; if all fail, it loads the last successful cached file. The file structure (multi-row header block) also varies across updates, so the parser tries multiple skip-row values.

The US Census API enforces a rate limit of approximately one request per 1.5 seconds across the full download (72 months). The pipeline inserts mandatory sleeps; running multiple instances simultaneously will cause 429 errors.

The CEPII servers are occasionally slow or offline. The pipeline sets a 120-second timeout per download attempt.

The Teti Global Tariff Database is access-restricted on Dropbox. The pipeline detects the access failure gracefully and proceeds without the product-level data, falling back to the Census country-month realized ETR. Researchers seeking access should contact Dr. Feodora Teti directly.

Fig. A. Data Pipeline Architecture

Notes: Four-layer reproducibility diagram of the data pipeline used to construct the analysis sample. Raw public sources are downloaded and cached, parsed and validated into intermediate files, merged into analysis panels, and used to generate every table and figure in the thesis. Arrows show the main data flow rather than every function-level dependency. Source: author.

APPENDIX B

Abbreviations and Acronyms

Every abbreviation used in the thesis is spelled out below, grouped by kind. Country codes follow the three-letter ISO 3166-1 alpha-3 standard; only the codes that appear as labels in Figures III and V are listed.

Term	Meaning
<i>Organizations and data sources</i>	
ACR	Arkolakis, Costinot, and Rodríguez-Clare (gains-from-trade formula)
CBP	United States Customs and Border Protection
CEPII	Centre d'Études Prospectives et d'Informations Internationales (French research centre in international economics)
CSMS	Cargo Systems Messaging Service (CBP trade-bulletin system)
IMF	International Monetary Fund
IMTS	International Merchandise Trade Statistics (IMF)
OECD	Organisation for Economic Co-operation and Development
QNA	Quarterly National Accounts (OECD)
USTR	Office of the United States Trade Representative
WEO	World Economic Outlook (IMF)
WTO	World Trade Organization
<i>Trade policy and law</i>	
EO	Executive Order
EU	European Union
FR	Federal Register
FTA	Free trade agreement
HS	Harmonized System (tariff nomenclature)
HS2	Harmonized System two-digit chapter
HTSUS	Harmonized Tariff Schedule of the United States
IEEPA	International Emergency Economic Powers Act
MFN	Most favoured nation (tariff rate)
USMCA	United States–Mexico–Canada Agreement
<i>Economics and econometrics</i>	
CES	Constant elasticity of substitution
CIF	Cost, insurance, and freight (import valuation)
DiD	Difference-in-differences
ETR	Realized effective tariff rate (duties collected divided by import value)
FE	Fixed effects
GDP	Gross domestic product
GLM	Generalized linear model
HC1	Heteroskedasticity-consistent covariance estimator (type 1)
JEL	Journal of Economic Literature classification codes (here F13, F14, C23)
OLS	Ordinary least squares
PPML	Poisson pseudo-maximum-likelihood
Q1, Q3	First and third calendar quarters
R	Market-share reallocation index (Equation 6)
SD	Standard deviation

Term	Meaning
SE	Standard error
S&P 500	Standard and Poor's 500 stock index
<i>Data and technical terms</i>	
API	Application programming interface
CAL_DUT_M0	Census variable: calculated duties, monthly
CON_VAL_M0	Census variable: imports for consumption value, monthly
CSV	Comma-separated values
CTY	Census country code
GEN_VAL_M0	Census variable: general imports value, monthly
ISO 3	Three-letter country codes (International Organization for Standardization)
JSON	JavaScript Object Notation
REST	Representational state transfer (API style)
SDMX	Statistical Data and Metadata eXchange
SHA-256	Secure Hash Algorithm, 256-bit
TMG_CIF_USD	IMF indicator: total merchandise goods, CIF, US dollars
URL	Uniform resource locator
USD	United States dollar
UTF-16	Unicode Transformation Format, 16-bit
XLS, XLSX	Microsoft Excel spreadsheet formats
ZIP	Compressed archive format
<i>Partner country codes (ISO 3) in Figures VI and VII</i>	
CAN	Canada
CHE	Switzerland
CHN	China
DEU	Germany
FRA	France
GBR	United Kingdom
HKG	Hong Kong
IDN	Indonesia
IND	India
IRL	Ireland
ITA	Italy
JPN	Japan
KHM	Cambodia
KOR	South Korea
MEX	Mexico
MYS	Malaysia
PHL	Philippines
SGP	Singapore
SVN	Slovenia
THA	Thailand
TWN	Taiwan
USA, US	United States
VNM	Vietnam

Term	Meaning
ZAF	South Africa

APPENDIX C

Reproducibility: Product-Exposure and Source-Validation Procedures

This appendix records the two procedures that extend the country-level panel: the product-level overlap measure and the independent source validation. Both are scripted so that a reader with the same inputs can reproduce every number. The scripts live in `scripts/` and read and write the same `data/` and `outputs/` folders as the main pipeline.

C.1 Census HS6 download and product-exposure measure

The country-level panel shows that China lost US market share while Taiwan, Vietnam, and others gained it, but it cannot show whether the gainers moved into the same products China had dominated. The product-exposure measure answers that by going to the six-digit Harmonized System (HS6) level.

1. *Get a Census API key.* Request a free key at `api.census.gov/data/key_signup.html` with a university email. Store it as the environment variable `CENSUS_API_KEY`, or in a local `census_data_api.txt` file as `CENSUS_API_KEY=...`; the download script reads either and the key is never committed to the repository.
2. *Identify the partners.* `scripts/download_hs6.py` derives the top 20 partners from the processed realized-ETR file `data/processed/realized_etr_by_country_month.csv`, ranked by total import value. It keeps only genuine partner codes (four digits, no leading zero, no letters), which drops the Census aggregate rows such as TOTAL FOR ALL COUNTRIES, OECD, and ASIA that would otherwise top the ranking.
3. *Download the detail.* For each partner and each month from January 2024 to February 2026 the script queries `api.census.gov/data/timeseries/intltrade/imports/hs` with `COMM_LVL=HS6`, retrieving the HS6 code (`I_COMMODITY`), import value (`GEN_VAL_MO`), duties (`CAL_DUT_MO`), and quantity. It respects the Census limit of roughly one request per 1.5 seconds, caches each partner separately so an interrupted run resumes where it stopped, and writes `data/raw/census_hs6_imports.parquet`. Run it with `python scripts/download_hs6.py`; the full pull takes about half an hour.
4. *Build the exposure measure.* `scripts/product_exposure.py` treats all months through March 2025 as the pre-shock window. Within it, for every HS6 product, the script computes China's share of US imports across the 20 partners and flags products where that share exceeds 30 percent as China-dominated. For each partner it then reports the fraction of its own pre-shock US exports that fell in those products, writing `outputs/product_exposure.csv`. Run it with `python scripts/product_exposure.py`.

The interpretation is in Section 5: a partner that gained US market share and also had high pre-shock overlap with China is exporting into the gap China vacated, the product-level fingerprint of rerouting that country-level data alone cannot reveal.

C.2 USITC DataWeb cross-source validation

The goal is to confirm that the Census import figures are not an artifact of a single collection system by comparing them with US International Trade Commission DataWeb.

1. *Export the second source.* From `dataweb.usitc.gov`, run an Imports for Consumption query reporting Customs Value, monthly, for 2025, across all countries, and export it to a spreadsheet. The export used here is stored at `data/raw/usitc_dataweb_2025.xlsx`.
2. *Reshape and compare.* `scripts/validate_usitc.py` reshapes the wide DataWeb sheet to one row per country-month, restricts both sources to the first and third quarters of 2025, aligns partner names across the two systems (for example Census "KOREA, SOUTH" to DataWeb "South Korea"), and computes the absolute percentage difference per partner-quarter. It writes `outputs/usitc_census_discrepancy.csv`. Run it with `python scripts/validate_usitc.py`.

The result is reported in Section 6: across the top 20 partners the two sources agree to a mean of 1.9 percent and a maximum of 7.7 percent in both quarters, the small residual reflecting the gap between general import value and customs value.

APPENDIX D

*

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APPENDIX E

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Tables

Table A. Summary Statistics

Variable	<i>N</i>	Mean	SD	Min	P25	Median	P75	P95	Max
<i>Bilateral trade flows (IMF IMTS panel, 2020–2025)</i>									
Trade flow (USD million)	72,621	4,109.6	19,735.7	0.00	6.0	72.0	746.1	15,944.7	357,666
ln(Trade flow)	71,281	4.27	3.18	-7.69	1.79	4.28	6.61	9.68	12.78
<i>Realized ETR (Census, country-month)</i>									
ETR, full sample (%)	17,838	2.10	4.45	0.00	0.08	0.63	2.24	9.88	200.00
ETR, 2020–2024 average (%)	14,860	1.39	2.34	0.00	0.07	0.50	1.58	5.69	200.00
ETR, 2025 (%)	2,978	5.68	6.92	0.00	0.27	3.67	9.03	17.47	70.35
<i>Gravity controls (CEPII)</i>									
Distance, distw (km)	61,726	9,504	3,412	1,854	n.a.	n.a.	n.a.	n.a.	16,466
Contiguous (0/1)	72,621	0.012	n.a.	0	n.a.	n.a.	n.a.	n.a.	1
Common official language (0/1)	72,621	0.311	n.a.	0	n.a.	n.a.	n.a.	n.a.	1
Colonial ties (0/1)	72,621	0.033	n.a.	0	n.a.	n.a.	n.a.	n.a.	1
Active FTA/WTO (0/1)	72,621	0.100	n.a.	0	n.a.	n.a.	n.a.	n.a.	1

Notes: Panel covers 235 partner countries and 73 monthly periods (January 2020 through January 2026). The realized ETR is the ratio of customs duties collected to general import value, computed at the country-month level from US Census Bureau administrative records. The 2025 mean (5.68 percent) is 4.1 times the 2020–2024 average (1.39 percent), reflecting the Liberation Day tariff increase. Values above 200 percent in the raw data are capped at 2.0 (see Section 4). Trade flow zeros and near-zeros (1,340 observations, 1.8 percent of the panel) are retained; PPML accommodates zero flows without a sample restriction. The percentile columns show how skewed the trade and tariff data are: the median trade flow (USD 72 million) is far below the mean (USD 4.1 billion), and the 2025 median ETR (3.67 percent) sits well below its mean (5.68 percent), so percentiles describe the typical partner better than the mean. Percentiles are shown only for the continuous outcome and treatment variables.

Source: IMF IMTS (International Monetary Fund 2025a), US Census Bureau (U.S. Census Bureau 2025), CEPII GeoDist (CEPII 2011), CEPII Gravity (CEPII 2022).

Table B. HS2 Sector Tariff Status, Liberation Day 2025

HS2	Description	Status	Authority	Rate (%)
27	Mineral fuels and energy	Exempt	Energy exemption	0
30	Pharmaceuticals	Exempt	Pharma exemption	0
62	Apparel (woven)	Tariffed	IEEPA baseline	10
61	Apparel (knit)	Tariffed	IEEPA baseline	10
64	Footwear	Tariffed	IEEPA baseline	10
72	Iron and steel	Tariffed	Section 232	25
73	Steel articles	Tariffed	Section 232	25
76	Aluminum	Tariffed	Section 232	25
84	Machinery, computers	Exempt	Tech exemption (CSMS)	0
85	Electronics, semiconductors	Exempt	Tech exemption (CSMS)	0
87	Vehicles and autos	Tariffed	Section 232 auto	25
90	Optical, medical instruments	Exempt	Tech exemption (CSMS)	0
94	Furniture	Tariffed	IEEPA baseline	10
95	Toys and games	Tariffed	IEEPA baseline	10

Notes: The electronics and semiconductor exclusions (HS 84, 85, 90) are documented in CBP CSMS message #64724565, which clarified the specified HTSUS headings and subheadings excluded from the reciprocal tariff and made them effective for merchandise entered for consumption on or after April 5, 2025 (importers use secondary classification 9903.01.32). The exclusion list is reproduced from that message in `data/raw/cbp_exemptions.csv`. USMCA-compliant goods from Canada and Mexico may be exempt regardless of chapter. Electronics (HS 85) is the largest US import category from China by value.

Source: USTR Annex I, April 2, 2025; Federal Register 90 FR 15,625; CBP CSMS #64724565; subsequent amendments through August 2025.

Table C. Trade Diversion Classification, Partner Countries by Response

Category	Definition	Countries
Double-winner	Exports to US up > 5% AND imports from China up > 5%	60
US diversion recipient	Exports to US up > 5%; China data missing or flat	13
Collateral damage	Both exports to US and imports from China down > 5%	15
Neutral	Neither threshold crossed	140
<i>Total classified</i>		228

Notes: Each country is classified by comparing average monthly trade values in Q1 2025 (pre-shock baseline) to Q3 2025 (post-shock window). Thresholds are ± 5 percent. Among the 88 countries that cross at least one threshold, the double-winner group is the largest category (60 of 88, 68 percent), consistent with a supply-chain rerouting story rather than pure substitution. The large neutral group reflects the short post-shock window and the limited universe of countries with meaningful pre-existing trade in both directions.

Source: IMF IMTS (International Monetary Fund 2025a); classification algorithm documented in the data pipeline (Ofłazoğlu, Çağan 2026).

Table D. Trade Diversion Classification: Threshold Robustness

Threshold	Double-winner	US diversion	Collateral	Neutral
<i>Panel A: country counts</i>				
$\pm 2\%$	70	9	18	131
$\pm 5\%$	60	13	15	140
$\pm 10\%$	48	20	11	149
<i>Panel B: share of pre-shock import value</i>				
$\pm 2\%$	0.062	0.000	0.000	0.937
$\pm 5\%$	0.029	0.001	0.000	0.970
$\pm 10\%$	0.023	0.006	0.000	0.970

Notes: Panel A repeats the classification of Table C at three thresholds. The double-winner group is the largest non-neutral category at every threshold. Panel B weights each category by its share of total pre-shock (Q1 2025) US import value: double-winners are many in number but account for a small fraction of import value, because the largest suppliers (China, Canada, Mexico) sit on category boundaries or in the neutral band. Source: IMF IMTS (International Monetary Fund 2025a) and author's calculations; generated by `compute_diversion_robustness` in the data pipeline (Ofłazoğlu, Çağan 2026).